Report on parameters of vgg16 and depthwise separable convolution

Parameters_conv2d = (kernel_height × kernel_width × input_channels) × output_channels + output_channels (biases)

Parameters_depth_conv2d = (kernel_height × kernel_width) × input_channels + input_channels (biases)

Parameters_dense = input_units × output_units + output_units (biases)

Parameters_batchnorm = 4 × channels (gamma, beta, moving_mean, moving_variance)

Parameters = 0(for pooling layer as it is not considered)

1. Vgg16
   (frozen+trainable)

Layer	Formula	Calculation	Params
Block1 Conv1	3×3×3×64 + 64	1,728 + 64	1,792
Block1 Conv2	3×3×64×64 + 64	36,864 + 64	36,928
Block2 Conv1	3×3×64×128 + 128	73,728 + 128	73,856
Block2 Conv2	3×3×128×128 + 128	147,456 + 128	147,584
Block3 Conv1	3×3×128×256 + 256	294,912 + 256	295,168
Block3 Conv2	3×3×256×256 + 256	589,824 + 256	590,080
Block3 Conv3	3×3×256×256 + 256	589,824 + 256	590,080
Block4 Conv1	3×3×256×512 + 512	1,179,648 + 512	1,180,160
Block4 Conv2	3×3×512×512 + 512	2,359,296 + 512	2,359,808
Block4 Conv3	3×3×512×512 + 512	2,359,296 + 512	2,359,808
Block5 Conv1	3×3×512×512 + 512	2,359,296 + 512	2,359,808

Layer	Formula	Calculation	Params
Block5 Conv2	3×3×512×512 + 512	2,359,296 + 512	2,359,808
Block5 Conv3	3×3×512×512 + 512	2,359,296 + 512	2,359,808
VGG16 Total			**14,714,688**

Custom trainable layers

Layer	Formula	Calculation	Params
GlobalAvgPool2D	-	-	0
Dense 1	512 × 512 + 512	262,144 + 512	262,656
Dropout	-	-	0
Dense 2	512 × 256 + 256	131,072 + 256	131,328
Dropout	-	-	0
Dense 3 (output)	256 × 3 + 3	768 + 3	771
Top Layers Total			**394,755**

- Non-trainable (frozen VGG16 layers) = 7,635,264
- Trainable (last 4 conv layers) = 7,079,424
- Trainable (top Dense layers) = 394,755
- TOTAL = 7,635,264 + 7,079,424 + 394,755 = 15,109,443

Optimizer params = 2 × 7,474,179 = 14,948,358

Total with optimizer = 7,635,264 + 7,474,179 + 14,948,358 = 30,057,801

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Total params: 30,057,803 (114.66 MB)
Trainable params: 7,474,179 (28.51 MB)
Non-trainable params: 7,635,264 (29.13 MB)
Optimizer params: 14,948,360 (57.02 MB)
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2. Depthwise separable convolution

Layer	Input Shape	Kernel	Formula	Params
Block 1				
DepthwiseConv2D	224×224×3	3×3	(3×3×3)	27
Conv2D (Pointwise)	224×224×3	1×1, 64 filters	(1×1×3)×64 + 64	192 + 64 = 256
BatchNormalization	224×224×64	-	4 × 64	256
MaxPooling2D	224×224×64	2×2	-	0

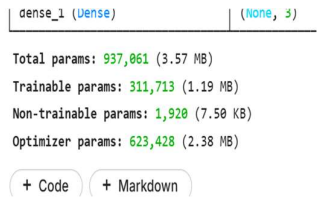
Layer	Input Shape	Kernel	Formula	Params
Block 2				
DepthwiseConv2D	112×112×64	3×3	(3×3×64)	576
Conv2D (Pointwise)	112×112×64	1×1, 128 filters	(1×1×64)×128 + 128	8,192 + 128 = 8,320
BatchNormalization	112×112×128	-	4 × 128	512
MaxPooling2D	112×112×128	2×2	-	0
Block 3				
DepthwiseConv2D	56×56×128	3×3	(3×3×128)	1,152
Conv2D (Pointwise)	56×56×128	1×1,	(1×1×128)×256 + 256	32,768 + 256

Layer	Input Shape	Kernel	Formula	Params
		256 filters		= 33,024
BatchNormalization	56×56×256	-	4 × 256	1,024
MaxPooling2D	56×56×256	2×2	-	0
Block 4				
DepthwiseConv2D	28×28×256	3×3	(3×3×256)	2,304
Conv2D (Pointwise)	28×28×256	1×1, 512 filters	(1×1×256)×512 + 512	131,072 + 512 = 131,584

Layer	Input Shape	Kernel	Formula	Params
BatchNormalization	28×28×512	-	4 × 512	2,048
Top Layers				
GlobalAveragePooling2D	28×28×512	-	-	0
Dense	512 → 256	256 units	(512×256) + 256	131,072 + 256 = 131,328
Dropout (0.5)	256	-	-	0
Dense	256 → 3	3 units	(256×3) + 3	768 + 3 = 771

Optimizer parameters = 2 × Trainable parameters
= 2 × 311,713
= 623,426

Total Trainable - 311,262



Total Non-trainable - 1,920

Model Total - 313,182